KABASINGA Arsene

SECOND
THIRD ITERATION

$m = 1.26 \quad b = 1.90$ Use points $(1,3), (3,6)$
$\alpha = 0.1$

For $(1,3)$ $\hat{y}_1 = (1.26)(1) + 1.90 = 3.16$

For $(3,6)$ $\hat{y}_2 = (1.26)3 + 1.90 = 5.68$

The predicted values are $(3.16, 5.68)$

Error $= y - \hat{y} \Rightarrow \begin{cases} 3 - 3.16 = -0.16 \\ 6 - 5.68 = 0.32 \end{cases} \Rightarrow (-0.16, 0.32)$

$$\frac{\Delta E}{\Delta m} = \frac{-2}{n} \sum (y - y_i) x_i \Leftrightarrow -\frac{2}{2}\left(1(3-3.16) + 3(6-5.68)\right)$$
$$= -1(-0.16 + 0.96) = -0.80$$

$$\frac{\Delta E}{b} = \frac{-2}{n} \sum (y - y_i) \Leftrightarrow -\frac{2}{2}\left((3-3.16) + (6-5.68)\right)$$
$$= -1(-0.16 + 0.32) = -0.16$$

So The updated m and b

$$M_{new} = M_{old} - \alpha \frac{\Delta E}{\Delta m} \Leftrightarrow (1.26) - (0.1)(0.80) = 1.34$$

$$B_{new} = B_{old} - \alpha \frac{\Delta E}{\Delta b} \Leftrightarrow (1.90) - (0.1)(-0.16) = 1.916$$

$M_{new} = 1.34$
$B_{new} = 1.916$

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